

THE VERMIGOLD FARMER'S ALMANAC · 2026 · FULL EDITION

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VERMIGOLD SOIL SYSTEMS

THE
**VermiGold
Farmer's Almanac**

A Regenerative Soil Guide for the High Desert

CEDAR CITY · ENOCH · ST. GEORGE

2024 EDITION



The VermiGold Farmer's Almanac

A Regenerative Soil Guide for the High Desert

2026 EDITION

*The desert is not a problem to be solved.
It is a system to be understood.*

PUBLISHED BY VERMIGOLD SOIL SYSTEMS · CEDAR CITY, UTAH

Dedicated to the growers who refused to give up on this ground.

The VermiGold Farmer's Almanac — A Regenerative Soil Guide for the High Desert, 2026 Edition.

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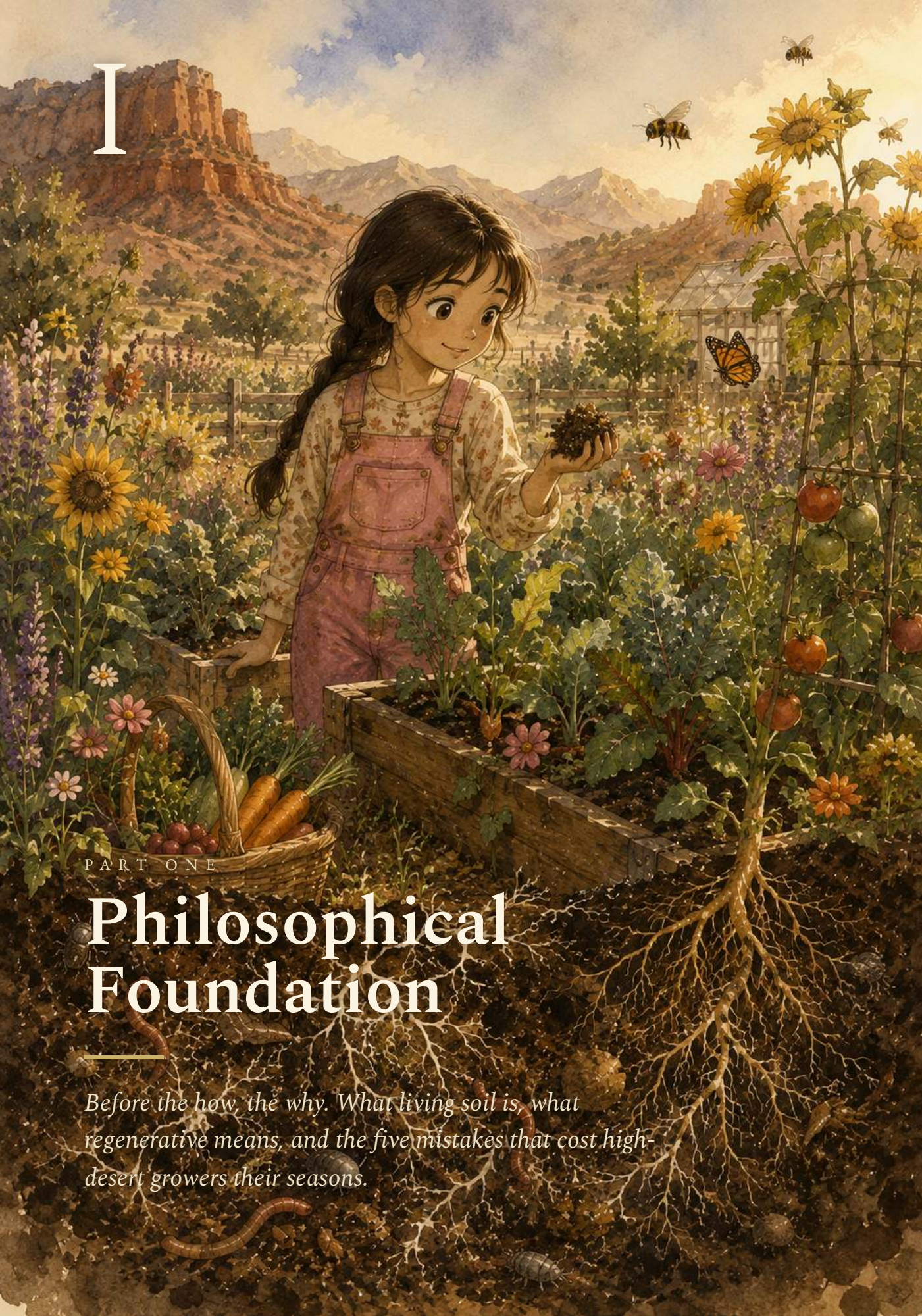
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I

PART ONE

Philosophical Foundation

Before the how, the why. What living soil is, what regenerative means, and the five mistakes that cost high-desert growers their seasons.

PART ONE · A LETTER

A Letter to the Grower

JEFFREY JENSON · VERMIGOLD SOIL SYSTEMS

If you've opened this almanac, you've probably had at least one season where the dirt won. The wind tore through your beds in May. The clay set up like concrete in July. The deer pruned your tomatoes overnight, and the yields were a fraction of what the seed packet promised.

We've been there. Most growers in southern Utah have. This book is not a list of garden tricks — it is a soil system playbook, built for the climate that runs from Cedar's frost-locked winters down through Enoch's hardpan and into St. George's furnace summers.

You will not find a chapter telling you to add 10-10-10 in spring. You will find chapters on the underground livestock you didn't know you owned — and how to feed them so they feed you back.

This almanac is not a book. It is a twelve-month system. Three stages. One piece of land that compounds in your favor every year you run it. Read in order if regenerative growing is new to you; skip to the playbook if you are already in season.

*Soil is a living system. When you fix the system,
everything improves.*

The land is not against you. It has been waiting for someone to listen. You are not buying products — you are entering a system. Welcome.

— *The VermiGold Team, Cedar City*

PART ONE

Why This Almanac Exists

Most growing guides are written for somewhere with rainfall. Southern Utah averages 8 to 13 inches of precipitation a year, and most of it lands when nothing is growing. Conventional advice does not survive here — and neither do most plants, when treated like they're growing in Iowa.

OUR SOILS, TYPICALLY

High pH — **7.8 to 8.4**

Calcium-rich, magnesium-poor

Organic matter often **under 1%**

A brick when wet, a dust when dry



PLATE II Dead dirt sheds water; living soil drinks it.

THREE THINGS DRIVE THIS BOOK

Regional honesty

Cedar at 5,800 ft and St. George at 2,600 ft are different climates. We treat them differently throughout.

System-first

A symptom is rarely the problem. We trace every issue to biology, water, and structure — then fix the system.

Long horizons

Recommendations build for next year, not just for Saturday.

If you want a quick fix, this is the wrong book. If you want to build a piece of land, read on.

PART ONE · THE LIVING SOIL PRINCIPLE

The Underground Livestock

Beneath every healthy bed runs a herd you never feed by hand. Bacteria, fungi, protozoa, nematodes, and arthropods cycle nutrients, build structure, and trade sugars with roots. Feed this food web and it feeds your plants — starve it, and no fertilizer can stand in for what it does. A bag of synthetic fertilizer is a feeding tube; it bypasses a healthy gut, and once you depend on it the gut shuts down.

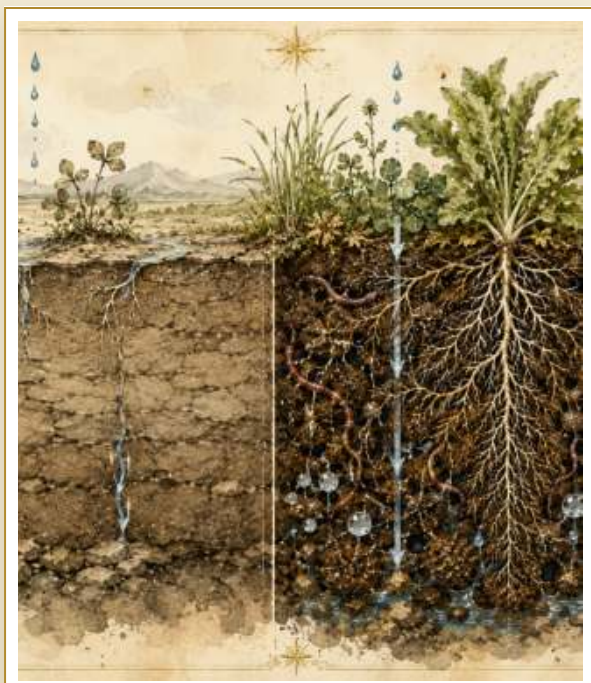


PLATE III *The soil food web, magnified. One teaspoon holds more organisms than people on Earth.*

THE FOOD WEB IN BRIEF

Bacteria & fungi break raw organic matter into plant-ready minerals and bind particles into crumb structure.

Protozoa & nematodes graze the microbes and release nitrogen exactly where roots can drink it.

Worms & arthropods shred, mix, and channel — turning dirt into living, breathing soil.

Feed the biology, and the biology will feed you. Everything else in this almanac is implementation.

PART ONE

The Regenerative Mindset

Regenerative is not a label. It is a direction. A piece of land is either getting healthier each year or getting worse — there is no neutral. The mindset shift is to ask, every season:

“Did I leave this ground better than I found it?”

FIVE PRINCIPLES

- 1 Minimize disturbance.** Every till and blast of synthetic salt resets the biological clock. Disturb only when there is no other option.
- 2 Keep something living.** A living root, twelve months a year, is the closest thing soil has to a heart. When the heart stops, the system stops.
- 3 Cover the surface.** Bare desert soil hits 150°F in July; microbes die at 130°F. Two inches of mulch can drop surface temperature 30–50 degrees.
- 4 Diversify.** Monocultures starve the biology. Mixed plantings, cover-crop blends, and rotation feed the full microbial community.
- 5 Integrate animals.** Worms, chickens, sheep, ducks. Animals close the loop and accelerate the cycle by years.

You will not implement all five at once. Pick one, run it for a season, layer the next. This is not a religion — it is a hierarchy of leverage.

PART ONE

What Most Growers Get Wrong

Five mistakes account for nearly every failed garden in the high desert — each one imported from somewhere wetter, cooler, or both.

MISTAKE 1 — TREATING SOIL LIKE A SUBSTRATE, NOT A SYSTEM

Nutrients without biology can't be absorbed in high-pH soils. **Fix:** feed the microbes; the microbes feed the plant.

MISTAKE 2 — WATERING SHALLOW AND OFTEN

Shallow watering trains shallow roots; the first heat dome kills the plant. **Fix:** water deeply, less often.

MISTAKE 3 — TILLING EACH SPRING TO “WAKE THE SOIL”

Tilling slices fungal networks and triggers a weed-seed explosion. **Fix:** top-dress on the surface; let worms pull it down.

MISTAKE 4 — BUYING SOLUTIONS ONE SYMPTOM AT A TIME

A fertilizer for yellowing, a spray for aphids — whack-a-mole while the system collapses. **Fix:** build the soil; symptoms quiet within two seasons.

MISTAKE 5 — SKIPPING THE OFF-SEASON

Bare beds all winter starve the biology; you restart from zero every May. **Fix:** cover crops, mulch, and a living root all twelve months.

If even one of these described you, you are not behind. You are ready.

VermiGold™

The G.R.O.W.T.H. System™

— A Living Soil Journal —

Biology
Awakens



Roots
Expand



Soil
Covered



Water
Stored



Ecosystem
Thrives



Legacy
Grows



Healthy soil isn't built in a day.
It's grown through cycles.



PART TWO

The Framework

The G.R.O.W.T.H. System™ is the operating framework of this almanac. It compresses a decade of regenerative field research into six leverage points any grower can run, season after season.

LETTER	PRINCIPLE	CORE ACTION
G	Grow Biology	Build the living workforce
R	Root Continuity	Never go fallow
O	Organic Cover	Never go bare
W	Water Retention	Slow, spread, sink every drop
T	Total Ecosystem	Above and below as one
H	Human + Soil Harmony	Work the land's rhythm

Each letter is a chapter, and every chapter follows the same structure: **what it is**, **why it matters**, **how to apply it**, and **what to look for** at 30, 90, and 365 days. Run the letters in order if you are starting from scratch; run them in parallel if your operation is established. They reinforce each other — biology feeds roots, roots feed structure, structure holds water, water grows ecosystem.



THE FIRST PRACTICE

Grow Biology

Build the living workforce.

WHAT IT IS

Build a thriving microbial workforce — bacteria, fungi, protozoa, nematodes, earthworms — that does the heavy lifting of plant nutrition for you.

WHY IT MATTERS

A square foot of biologically rich soil delivers nitrogen, phosphorus, and trace minerals on demand — and microbes make the glues that bind dust into structure.

HOW TO APPLY IT IN THE HIGH DESERT

- ◆ **Inoculate.** Apply a high-biology amendment once soil hits 50°F in early spring.
- ◆ **Stop killing it.** Avoid synthetic nitrogen, broad fungicides, and tillage above 120°F.
- ◆ **Feed it.** Top-dress finished compost or vermicast ¼–½" in spring and fall.
- ◆ **Diversify the menu.** Plant four to six species in any area each season.

WHAT TO LOOK FOR

30 days

Visible earthworm activity in the top six inches.

90 days

A moist handful crumbles into aggregates, never smears.

365 days

Organic matter up 0.3–0.8% — a generational gain in one season.

When biology is alive, you stop fertilizing. You start feeding.

G GROW BIOLOGY · THE METHODS

Ways to Wake the Workforce



GROW BIOLOGY *Five ways to put a living microbial workforce back into high-desert ground.*

Liquid microbial drench

FASTEST INOCULATION

VermiGold extract delivers the full consortium straight to the root zone. 1:32, every application month.

Compost & vermicast

FOOD + SLOW BIOLOGY

Top-dress ¼–½" on the surface in spring and fall. Worms carry it down.

Compost tea / extract

AMPLIFY BETWEEN DRENCHES

Brewed castings at 1:10, applied as a foliar or soil drench to multiply biology fast.

Diversify the menu

FEED THE WHOLE WEB

Plant 4–6 species per area and mixed cover blends — different roots feed different microbes.

Biochar / terra-preta charge

HABITAT FOR MICROBES

Charged carbon gives biology somewhere to live in dead dirt — one application per quarter.

Layer the methods — drench to inoculate, compost to feed, diversity to sustain.

R

THE SECOND PRACTICE

Root Continuity

Never go fallow.

WHAT IT IS

Keep a living root in the soil twelve months a year — the closest thing your ground has to a beating heart.

WHY IT MATTERS

Roots pump liquid carbon into the rhizosphere all day, feeding the microbes that build structure. Fallow soil starves the biology within weeks and resets to year zero.

HOW TO APPLY IT IN THE HIGH DESERT

- ◆ **Winter cover.** Broadcast cereal rye, winter peas, and hairy vetch in October.
- ◆ **Successive planting.** Reseed within seven days of pulling a finished crop.

- ◆ **Summer cover.** St. George July beds run cowpea, sorghum-sudan, and buckwheat.
- ◆ **Perennials on the fence line.** Asparagus, rhubarb, native grasses, fruit trees.

WHAT TO LOOK FOR

30 days

Bare beds replaced with a green stand of cover.

90 days

Visibly darker soil under perennial root zones.

365 days

No more “starting from scratch” each spring. The system holds.

A piece of land with no living roots is a piece of land going backwards.

R ROOT CONTINUITY · THE METHODS

Ways to Keep Living Roots



ROOT CONTINUITY *Five ways to keep a living root in the ground twelve months a year.*

Winter cover blend

PLANT SEP–OCT

Cereal rye + winter peas + hairy vetch. Overwinters even in Cedar; terminate in April.

Summer cover blend

PLANT JUN

Cowpea + sorghum-sudan + buckwheat keep roots alive through St. George heat dormancy.

Successive & relay planting

NEVER GO FALLOW

Reseed within seven days; interplant the next crop before the last comes out.

Perennial anchors

YEAR-ROUND ROOTS

Asparagus, rhubarb, fruit trees, and native grasses hold roots where annuals can't reach.

Quick-rotation blend

ROOTS IN 60 DAYS

Daikon + clover + oats punch compaction, fix nitrogen, and build biomass fast.

A living root is the soil's heartbeat. Choose the blend the season can carry.



THE THIRD PRACTICE

Organic Cover

Never go bare.

WHAT IT IS

Never let your soil surface go naked. Cover it with mulch, cover crops, residue, or a living canopy at all times.

WHY IT MATTERS

Bare desert soil in July reaches 140–160°F; microbes die at 130°F. By August the top inch of an uncovered bed is functionally sterile.

HOW TO APPLY IT IN THE HIGH DESERT

- ◆ **Mulch deep.** 2–4" of straw, leaves, chips, or alfalfa on every bed by mid-May.
- ◆ **Living mulch.** Underplant tomatoes and squash with low clovers.
- ◆ **Don't till residue under.** Lay it on top; let the worms pull it down.
- ◆ **In St. George, double up.** Compost + straw + drip saves 30–50% water.

WHAT TO LOOK FOR

30 days

Surface stays cool to the touch even at noon in July.

90 days

The layer below the mulch is dark, moist, and crumbly.

365 days

Mulch has decomposed into soil. Apply again — the cycle is the system.

A bed with bare soil is a bed losing money.

O ORGANIC COVER · THE METHODS

Ways to Cover the Soil



Sheet mulching

BEST FOR NEW BEDS

Cardboard on the weeds, wet it, then 4–6" compost + 3–4" straw. Plant into the compost — no tilling.

Loose organic mulch

ANNUAL BEDS & PATHS

Straw, shredded leaves, or wood chips, 2–4". Pull back a 2" ring from stems.

Living mulch

SHADE + NITROGEN

Underplant low clovers (Dutch white, crimson) beneath tall crops to shade soil while feeding it.

Cover-crop canopy

OFF-SEASON BEDS

A standing cover crop is its own living cover — roots below, canopy above.

Chop-and-drop residue

FREE, CONTINUOUS

Lay spent crops and prunings on the surface. Let the worms and microbes pull them down.

Bare soil is a wound. Any of these closes it — in St. George, layer two.



THE FOURTH PRACTICE

Water Retention

Slow, spread, sink every drop.

WHAT IT IS

Capture every drop that touches your land, slow it down, and store it in living soil rather than letting it run off or evaporate.

WHY IT MATTERS

The job isn't to add water — it's to keep what arrives. A 1% gain in organic matter holds an extra **16,000–25,000 gallons per acre**.

HOW TO APPLY IT IN THE HIGH DESERT

- ◆ **Build organic matter.** Compost, vermicast, mulch, cover crops — every layer raises capacity.
- ◆ **Shape the land.** Swales, berms, and basins on contour catch monsoon runoff.
- ◆ **Drip, don't spray.** Subsurface drip cuts evaporation 60% versus overhead.
- ◆ **Mulch first, water second.** A mulched bed needs roughly half the irrigation.

WHAT TO LOOK FOR

30 days

Irrigation runtime drops 20–40%.

90 days

Beds stay moist three to five days longer between waterings.

365 days

Water bill cut in half; plants hold steady at 105°F.

Water is the limiting nutrient in the desert. Build soil that holds it, and you have changed the equation.

W

WATER RETENTION · THE METHODS

Ways to Hold Every Drop



WATER RETENTION *Five ways to slow, spread, and sink the water you already get.*

Build organic matter

THE SOIL SPONGE

Every 1% gain holds 16,000+ extra gallons per acre. Compost, vermicast, mulch, cover.

Mulch blanket

FREE, IMMEDIATE

A 2" layer cuts evaporation 50–70% — the cheapest water you'll ever "buy."

Swales & berms on contour

CATCH THE MONSOON

Earthworks on contour catch runoff and force it to infiltrate before it leaves the property.

Roof & rain catchment

STORE THE FREE WATER

A 1,000 sq ft roof + 1" of rain = 600 gallons. Route gutters into growing zones.

Subsurface drip + timing

DELIVER IT DEEP

Subsurface drip cuts evaporation 60%. Run pre-dawn, deep and infrequent, to train roots down.

Don't add more water — keep the water already arriving.

T

THE FIFTH PRACTICE

Total Ecosystem

Above and below as one.

WHAT IT IS

Manage above-ground and below-ground as one integrated system. Pollinators, predators, birds, and livestock are not side projects — they are infrastructure.

WHY IT MATTERS

A garden that is only plants is fragile. One outbreak ends the season. A functional ecosystem absorbs shocks because every organism has a job.

HOW TO APPLY IT IN THE HIGH DESERT

◆ **Plant for pollinators.** Penstemon, blanket flower, milkweed, sunflower — April to October.

◆ **Run hens through beds.** They eat pests, scratch in residue, add fertility.

◆ **Build habitat.** Insect hotels and bare patches for ground-nesting native bees.

◆ **Stack functions.** Plan every plant for at least three jobs.

WHAT TO LOOK FOR

30 days

Visible increase in bee and beneficial insect traffic.

90 days

Aphid colonies disappear without intervention.

365 days

You no longer reach for sprays. The system runs itself.

A regenerative garden is a habitat that grows food — not a food plot that excludes everything else.

TOTAL ECOSYSTEM · THE METHODS

Ways to Build the Whole Web



TOTAL ECOSYSTEM *Five ways to turn a food plot into a functioning habitat.*

Pollinator strips

APRIL–OCTOBER BLOOM

Penstemon, milkweed, sunflower, calendula. Pollinators triple fruit set on cucurbits and trees.

Beneficial-insect habitat

BRING THE PREDATORS

Insect hotels and bare ground for ground-nesting native bees. Lacewings and wasps end aphids without sprays.

Integrate poultry & stock

CLOSE THE LOOP

Run hens through finished beds — they eat pests, scratch in residue, and add fertility.

Predator perches & boxes

RODENT CONTROL

Kestrel and owl boxes keep rodent pressure off your root crops, for free.

Stack functions

THREE JOBS EACH

A fruit tree is shade, windbreak, pollinator forage, and a harvest. Plan every plant for three jobs.

A garden that is only plants is fragile. A habitat absorbs the shocks.

H

THE SIXTH PRACTICE

Human + Soil Harmony

Work the land's rhythm.

WHAT IT IS

Work in rhythm with the land's biological clock — not the calendar's, not the seed catalog's, not the neighbor's.

WHY IT MATTERS

The biggest mistake is doing the right thing at the wrong time. Soil temperature, not air temperature, drives biology.

HOW TO APPLY IT IN THE HIGH DESERT

◆ **Buy a soil thermometer.** A \$15 tool prevents thousands in failed plantings.

◆ **Slow down in extremes.** July in St. George and January in Cedar are rest seasons.

◆ **Watch the indicator plants.** Rabbitbrush blooms, monsoon comes; cottonwoods drop, frost nears.

◆ **Journal weekly.** One observation a week becomes your most accurate forecast.

WHAT TO LOOK FOR

30 days

Timing decisions feel grounded, not anxious.

90 days

Fewer failed plantings, bolted greens, scorched seedlings.

365 days

The calendar in your head matches the calendar of the land.

The grower who is in rhythm with the land does less and gets more.

H HUMAN + SOIL HARMONY · THE METHODS

Ways to Read & Time the Land



HUMAN + SOIL *Five ways to work in rhythm with the land's biological clock.*

The soil-thermometer routine

A \$15 TOOL

Read at 2" deep, same time daily. Plant by soil temperature, not by date.

Indicator plants & phenology

READ THE WILD

Rabbitbrush bloom signals monsoon; cottonwoods dropping means frost is two weeks out.

The field journal

ONE NOTE A WEEK

Soil temp, rain, what's blooming, what's failing. By year five it beats any zone map.

The Vermigold slide rule

TEN-SECOND ANSWER

Four inputs — month, zone, crop, soil — return the action, method, and frequency.

Seasonal pacing

PUSH & REST

Work hard in the shoulder months; slow down in July's heat and January's cold.

Do the right thing at the right time, and the land does the rest.

PART TWO

G.R.O.W.T.H. as an Annual Rhythm

The six letters are not a checklist. They are a wheel that turns through the year. Your job is to give it a deliberate push at the right moment — not to spin it constantly.

LATE WINTER · FEB–MAR	G + H	Inoculate biology before it wakes. Prep the plan before the rush.
SPRING · APR–MAY	R + O	Establish living roots and lay down cover. Set the foundation.
EARLY SUMMER · JUN	W + T	Tighten water systems before peak heat. Bring on pollinators.
HIGH SUMMER · JUL–AUG	O+W+H	Defend the system. Mulch deeper, water smarter, harvest cool.
LATE SUMMER · SEP	G + R	Re-inoculate after heat stress. Plant fall cover and fall crops.
FALL · OCT–NOV	R+O+T	Lock in winter cover. Run animals. Stack residue.
WINTER · DEC–JAN	H	Rest. Plan. Sharpen tools. Read.

The wheel turns whether you turn it or not. Run it three years and the land is no longer the land you started with.

III

PART THREE

Regional Intelligence

Cedar at 5,800 feet and St. George at 2,600 feet are not the same climate. Know the ground beneath your feet.

PART THREE

High Desert Realities

The high desert is not a single climate. It is a stack of climates separated by elevation, exposure, and a few thousand feet of vertical drop. But five things are true everywhere in southern Utah.

Brutal evaporation

UV index above 8 for six months; wind regularly over 20 mph; summer ET exceeds 6 inches a month.

Calcareous, alkaline soils

pH 7.8–8.4 is normal. Iron and zinc lock out; phosphorus is present but unavailable without microbes.

Low organic matter

Most untouched land sits at 0.4–0.9%. Healthy growing soil is 4–6%. The gap is your work.

Volatile precipitation

Cedar averages 11", St. George 8". You may get 3" in 45 minutes, then nothing for 90 days.

Wide temperature swings

Cedar can hit 95°F at noon and 45°F at sunrise the same June day.

WHAT THIS MEANS FOR THE GROWER

Soil-building is not optional — it is the only way to buffer the volatility. Generic gardening advice will lose you a season. Mulch and biology are not extras; they are core infrastructure.

PART THREE · REGIONAL PROFILE

Cedar City & Enoch

5,800 FT · ZONE 6A–6B

CLIMATE

Frost-free days	110–130
Last spring frost	May 20–Jun 5
First fall frost	Sep 25–Oct 10
Precipitation	10–12"/yr
Summer highs	88–95°F
Winter lows	0–15°F

SOILS

Deep alluvial fan — calcareous loam over caliche at 12–36". Hardpan common; drainage variable. Native pH 7.9–8.3.

SEASONAL STRESS POINTS

Spring wind (Apr–May). Desiccates transplants in 24 hours. Use windbreaks.

Late frost surprise. Memorial Day frosts hit 1 year in 4. Hold heat crops to Jun 5.

Monsoon hail. Jul–Aug storms shred crops in 10 minutes. Keep row cover ready.

Early fall frost. Run season extension by Sep 10.

ADVANTAGES

Cool nights mean exceptional flavor in tomatoes, brassicas, alliums, and greens. Long winters give excellent stone-fruit dormancy. Lower pest pressure than warmer zones.

CEDAR’S CENTRAL PLAY

Front-load biology in late April, defend through the wind, and ride a long, productive June through early September.

PART THREE · REGIONAL PROFILE

St. George

2,600 FT · ZONE 8A–8B

CLIMATE

Frost-free days	220–260
Last spring frost	late Feb–mid Mar
First fall frost	Nov 10–25
Precipitation	7–9"/yr
Summer highs	100–110°F
Winter lows	25–35°F

SOILS

Red sandy loam over Moenkopi shale and limestone. Drains fast — sometimes too fast. OM often below 0.5%; pH 8.0–8.4. Lockout, not deficiency, is the issue.

SEASONAL STRESS POINTS

Heat dormancy (late Jun–Aug). Cool crops bolt. Treat summer as survival, not production.

UV burn. Even shade-tolerant plants need 30–40% shade cloth in July.

Salt accumulation. Hard water + ET concentrate salts. Deep-flush periodically.

Winter underestimation. 22°F nights can kill citrus and tender perennials.

ADVANTAGES

Two distinct growing seasons (Feb–Jun, Sep–Dec). Citrus, fig, pomegranate, jujube, and Mediterranean herbs thrive. Year-round greens with shade and cover.

ST. GEORGE’S CENTRAL PLAY

Plant heavy in February, rest in July, plant again in September. Treat the calendar like a Mediterranean climate, not a Midwest one.

PART THREE

Microclimate Mapping

Every property has hidden microclimates. Identifying them is worth more than any single amendment you will ever buy. Map five variables across a full year.

- 1 Sun exposure.** Track sun and shade at 9 AM, noon, and 4 PM in midsummer and midwinter. South walls add 8–15°F.
- 2 Wind exposure.** Walk it on a windy day. Quiet pockets get heat-lovers; exposed corners get windbreaks.
- 3 Cold-air drainage.** Cold air flows downhill like water. The low spot is 5–10°F colder on still nights — never plant blossoms in a frost pocket.
- 4 Water flow.** Walk it during a hard rain. Where does water move, pool, leave? Swales go on contour above the dry spots.
- 5 Soil variability.** Dig three to five test pits. Note depth to hardpan, color, structure. You may have two or three soils on a half-acre.

THE MAPPING EXERCISE

Take a printed satellite image of your property. Spend a year marking the five variables. By the end you have a custom climate map worth more than any USDA zone — and you plant the right thing in the right place every time.

IV

PART FOUR

The Month-by-Month Playbook

Twelve months, two regions, one operating manual for the year.

PART FOUR

How to Use the Playbook

The next twelve pages are the operating manual for the year. Every month follows the same four-part structure — read it the same way each time.

In the soil

The biology, temperature, and moisture state of the ground this month.

What to do

The two to four highest-leverage actions — a priority list, not a chore list.

What to avoid

The classic high-desert mistakes that cost growers whole seasons.

VermiGold window

When and how to apply — some months a heavy drench, some a light primer — but biology goes down every month.

READ THREE MONTHS AT A TIME

The grower who reads only the current month is reactive. Read the month behind and the month ahead.

USE THE MARGINS

Every page has room for your notes. By 2027 your annotations are as valuable as the printed text.

Two regions, one playbook. Where Cedar and St. George diverge, we call it out.

PART FOUR

Using the VermiGold Slide Rule

Bundled with the premium edition is a precision decision tool. The almanac teaches the *why*; the slide rule executes the *what*.

Four inputs in. Three answers out. Under ten seconds.

YOU SET

Month — Jan to Dec

Climate zone — Cedar/Enoch or St. George

Crop type — veg, fruit trees, turf, cannabis

Soil condition — poor, average, healthy

THREE WINDOWS REVEAL

Action step — plant, build biology, protect, or hold

VermiGold method — root soak, soil spray, foliar, top dress

Frequency — weekly, bi-weekly, monthly, seasonal

THE THREE MOMENTS THE RULE EARNS ITS KEEP

- ◆ The first warm Saturday in spring, when you're tempted to plant too early.
- ◆ The hottest week of July, when you're tempted to water at midday.
- ◆ The first cool September morning, when most growers miss the year's most important application.

Spin. Align. Act.

January

WINTER REST

Deep rest. The year is planned from here.

DAYLIGHT 9.5 HRS · BIOLOGY DORMANT

CONDITION

CEDAR · ENOCH

ST. GEORGE

Soil temperature

Frozen 6–18"

38–48°F

Frost

Hard freeze

Light, rare

Water need

Minimal

Minimal

Cedar is frozen above the frost line and biology sits near zero. St. George idles, cool and slow.

DO THIS MONTH

BOTH

Map this year's beds, order seed, schedule deliveries.

CEDAR

Prune dormant fruit trees on still, dry days above 28°F.

BOTH

Sharpen and oil tools — a clean pruner is a healthy pruner.

ST. GEORGE

Plant bare-root trees and garlic; sow peas and spinach under low tunnels.

QUICK WIN

Pull a soil test now — labs are uncrowded, turnaround fast.

AVOID

Working wet or frozen ground — one afternoon compacts for years.

WATCH FOR

Keep all weight off saturated and frozen beds.

VERMIGOLD THIS MONTH — MAINTENANCE PRIMER



Both regions: a light biology primer keeps the microbial reserve alive through dormancy. 1:128, one pass on a mild day above 40°F. Then map your twelve-month schedule.

February

LATE WINTER

St. George wakes. Cedar still waits.

DAYLIGHT 10.5 HRS · BIOLOGY STIRRING

CONDITION	CEDAR · ENOCH	ST. GEORGE
Soil temperature	Locked	45–55°F
Frost	Locked	Last ~Feb 25
Water need	Low	Low

St. George soil crosses 45°F — the threshold where bacterial life rebounds. Cedar stays locked.

DO THIS MONTH

ST. GEORGE

Direct-sow peas, spinach, lettuce, kale, carrots, beets late month.

CEDAR

Start onions, leeks, brassicas, and herbs indoors under lights.

ST. GEORGE

Transplant onion sets; protect with low tunnels.

BOTH

Build or refresh compost — it will heat up by month's end.

QUICK WIN

Time St. George's first drench to the week roots go in.

AVOID

Cedar planting outside — hardy seed rots until soil hits 40°F.

WATCH FOR

Soil crossing 45°F is the go-signal in St. George.

VERMIGOLD THIS MONTH — DRENCH



St. George: first drench of the year, late month, on beds planted by Feb 25. 1:32, 1 qt / 25 sq ft. Cedar: hold.

March

WAKE-UP

The month the soil wakes up.

DAYLIGHT 12 HRS · BIOLOGY WAKING

CONDITION	CEDAR · ENOCH	ST. GEORGE
Soil temperature	35 → 50°F	50 → 60°F
Frost	Surprise frosts	Past
Water need	Low	Low-Mod

Cedar climbs from 35°F toward 50°F; St. George reaches 60°F. Microbial life accelerates fast.

DO THIS MONTH

CEDAR

Late month, direct-sow peas, spinach, radish if soil crumbles (squeeze test).

CEDAR

Start tomatoes and peppers indoors.

ST. GEORGE

Set out brassica transplants; direct-sow beans late.

BOTH

First compost / vermicast top-dress, ¼–½" on every bed.

QUICK WIN

The March foundation drench sets the season’s trajectory.

AVOID

Tilling to “wake” beds — it buries the biology already waking.

WATCH FOR

Squeeze test — soil should crumble, never smear.



VERMIGOLD THIS MONTH — FOUNDATION DRENCH

Both regions: foundation drench on every active bed. 1:32, 1 qt / 25 sq ft. The single most important application of the year.

April

Plant and protect.

WAKE-UP

DAYLIGHT 13 HRS · BIOLOGY ACTIVE

CONDITION

CEDAR · ENOCH

ST. GEORGE

Soil temperature

50–60°F

60–70°F

Frost

Wind is the killer

Past

Water need

Moderate

Moderate

Soil is warm enough for active growth region-wide — but the nights are still cold in Cedar, and the wind does the killing.

DO THIS MONTH

CEDAR

Sow brassicas, roots, hardy greens; hold tomatoes and peppers to May 15.

BOTH

Terminate cover crops to surface mulch; establish windbreaks.

ST. GEORGE

Sow beans, corn, squash; set out tomatoes and peppers; mulch 3–4".

BOTH

Plant pollinator strips — calendula, borage, alyssum.

QUICK WIN

Soak every root ball in VermiGold before it touches the ground.

AVOID

Bare soil under transplants.
Mulch the day you plant.

WATCH FOR

Spring wind dehydrates an exposed transplant in 24 hours.



VERMIGOLD THIS MONTH — SOAK

Both regions: transplant soak mid-month — dunk root balls before planting, then a second drench on transplant beds. Dilute 1:32.

May

Peak planting. The whole region is growing.

PEAK GROWTH

DAYLIGHT 14 HRS · BIOLOGY PEAK

CONDITION	CEDAR · ENOCH	ST. GEORGE
Soil temperature	60–70°F	70–80°F
Frost	Until ~Jun 5	None
Water need	High	High

Peak biological activity. Soil temperatures climb fast toward summer; the last reliable spring rains are passing.

DO THIS MONTH

CEDAR

After May 20, set out tomatoes, peppers, eggplant, melon, squash.

BOTH

Mulch every square foot, 3–4" (4–6" in St. George).

ST. GEORGE

Last warm-season window before heat lockout — everything in by mid-May.

BOTH

Test and tighten drip lines before peak heat.

QUICK WIN

A sunrise foliar coat now heads off mid-summer disease.

AVOID

Midday overhead watering — you feed the air, not the plant.

WATCH FOR

Cedar can still frost until about June 5 — keep cover ready.



VERMIGOLD THIS MONTH — FOLIAR

Both regions: foliar spray early month. 1:64, at sunrise or sunset. Builds disease resistance and leaf-surface microbes.

June

Water is the limiting factor.

PEAK GROWTH

DAYLIGHT 14.5 HRS · BIOLOGY HIGH

CONDITION

CEDAR · ENOCH

ST. GEORGE

Soil temperature

65–75°F

80–95°F

Frost

None

None

Water need

High

Very High

Soil is hot and getting hotter, especially at the St. George surface. Roots need depth and shade.

DO THIS MONTH

CEDAR

Stake tomatoes, prune for airflow, side-dress with compost.

BOTH

Begin succession planting — beans and lettuce every two weeks.

ST. GEORGE

Install 30–40% shade cloth over greens; plan for July dormancy.

BOTH

Harvest garlic at scape time, then again at brown-leaf time.

QUICK WIN

Cover bare patches immediately as the surface heats.

AVOID

Heavy nitrogen feeds — lush growth wilts in July heat.

WATCH FOR

Monitor soil temperature daily; the surface is climbing fast.

VERMIGOLD THIS MONTH — DRENCH



Cedar: continue monthly drench mid-month, 1:32. St. George: pre-dormancy support drench, paired with deep watering and shade cloth.

July

HIGH SUMMER

The hardest month. Defend the system.

DAYLIGHT 14 HRS · BACTERIA DOMINANT

CONDITION	CEDAR · ENOCH	ST. GEORGE
Surface temperature	110–120°F	140°F+
Frost	None	None
Water need	High	Peak

Surface soils bake; mulched and shaded beds run 30–50°F cooler. Mycorrhizal activity slows, bacteria dominate, and monsoon storms begin mid-month.

DO THIS MONTH

CEDAR

Peak harvest of tomatoes, beans, squash; watch for monsoon hail.

BOTH

Deepen mulch to 4"; re-mulch any thin spots immediately.

ST. GEORGE

Survival mode — maintain only what's established; harvest early morning only.

BOTH

Capture monsoon water into swales, basins, and rain barrels.

QUICK WIN

Hand-water young trees deeply once a week, even with drip running.

AVOID

Working in the heat of day — care 6–9 AM and 7–9 PM only.

WATCH FOR

First monsoon hail — keep floating row cover on standby.



VERMIGOLD THIS MONTH — FOLIAR ONLY

Cedar: light foliar (1:64) at sunset on stressed plants; hold drench until soil cools below 90°F. St. George: hold drench; early-morning foliar mist (1:128).

August

Recover. Begin the fall.

LATE SUMMER

DAYLIGHT 13 HRS · BIOLOGY REBOUNDED

CONDITION	CEDAR · ENOCH	ST. GEORGE
Soil temperature	Warm	Hot, flushing
Frost	None	None
Water need	High	High

Monsoon moisture brings sudden flushes of microbial activity, then evaporation pulls it back. Well-prepared beds keep producing; unprepared beds crash.

DO THIS MONTH

CEDAR

Begin fall planting late month: brassicas, lettuce, spinach, beets, carrots in shade.

BOTH

Capture every monsoon storm; direct gutters into growing zones.

ST. GEORGE

Plan the fall season; order seed; repair sun-damaged drip lines.

BOTH

Top-dress compost on any bed where production is dropping.

QUICK WIN

Reseed within seven days of pulling a spent crop.

AVOID

Waiting until September for fall crops in Cedar — you'll run out of season.

WATCH FOR

Tomato and pepper disease — increase airflow, remove affected leaves.

VERMIGOLD THIS MONTH — RECOVERY DRENCH



Both regions: recovery drench late month, 1:32, after the first cool night below 65°F. This restarts biology after heat dormancy.

September

The second spring.

FALL PLANTING

DAYLIGHT 12 HRS · BIOLOGY REBOUNDING

CONDITION	CEDAR · ENOCH	ST. GEORGE
Soil temperature	55–65°F	70–80°F
Frost	Sep 15 +	None
Water need	Moderate	Moderate

Soil is cooling and microbial activity rebounds. Fall is the second spring — in many ways more productive than the first for cool-season crops.

DO THIS MONTH

CEDAR

Direct-sow all cool-season crops by Sep 5; plant garlic late; start fall cover.

BOTH

Side-dress active beds with vermicast or finished compost.

ST. GEORGE

Heavy fall planting: brassicas, greens, roots; prep beds for a second tomato crop.

BOTH

Run hens through finished beds to eat residue and add fertility.

QUICK WIN

The first cool morning is the cue for the year’s most important application.

AVOID

Delaying fall planting — the cool-season window is shorter than it seems.

WATCH FOR

Early frost surprise in Cedar after Sep 15 — keep cover ready.

VERMIGOLD THIS MONTH — MAJOR FALL DRENCH



Both regions: major fall drench, 1:32. The second-most important application of the year — it sets the soil’s biological reserves for winter.

October

FALL & FROST

Lock in the winter.

DAYLIGHT 11 HRS · FUNGI TAKE OVER

CONDITION

CEDAR · ENOCH

ST. GEORGE

Soil temperature

Into the 40s°F

60s°F

Frost

Oct 5–15

Not yet

Water need

Low–Mod

Moderate

Bacteria slow; fungi take over. Decomposition of summer residue accelerates. Frost arrives in Cedar mid-month; St. George has four to six weeks yet.

DO THIS MONTH

CEDAR

Cover or harvest tender crops; plant garlic; sow cover on every empty bed.

BOTH

Build leaf piles; run them through a mower for faster mulch.

ST. GEORGE

Continue heavy fall production; plant garlic, onions, second greens.

BOTH

Lay fall mulch 3–4" deep on beds going into winter.

QUICK WIN

October garlic for July harvest — a top-yield decision all year.

AVOID

Pulling crops and leaving beds bare for winter — the worst single move.

WATCH FOR

Don't miss the leaf window — by November they're gone.

VERMIGOLD THIS MONTH — FINAL / CONTINUE



Cedar: final drench of the year before soil drops below 50°F. St. George: continue monthly drench, 1:32.

November

SHUT DOWN

Shut down in Cedar. Produce in St. George. DAYLIGHT 10 HRS · BIOLOGY SLOWING

CONDITION	CEDAR · ENOCH	ST. GEORGE
Soil temperature	→ freezing	50–60°F
Frost	Hard freeze	Nov 10–25
Water need	Low	Low

Cedar soils cool toward freezing; fungal networks stay active under mulch. St. George holds in productive 50–60°F territory.

DO THIS MONTH

CEDAR Final harvest of stored crops; mulch all beds; bring herbs to protection.	ST. GEORGE Cover tender crops at first frost; harvest citrus through the month.
BOTH Inventory tools; drain and store hoses; insulate hose bibs.	BOTH Build cold frames or low tunnels for winter greens.

QUICK WIN Drain irrigation lines — a frozen split is a \$400 spring repair.	AVOID Working soggy soil after the first hard rain or snow.	WATCH FOR Cover crops are doing their best work now — don’t disturb them.
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VERMIGOLD THIS MONTH — MAINTENANCE  Cedar: light biology primer before the deep cold, 1:128. St. George: maintenance drench mid-month, 1:64.

December

WINTER REST

Deep rest. Next year starts now.

DAYLIGHT 9.5 HRS · BIOLOGY DORMANT

CONDITION	CEDAR · ENOCH	ST. GEORGE
Soil temperature	Dormant	45–55°F
Frost	Hard freeze	22–35°F
Water need	Minimal	Low

Cedar is dormant; snow is the best insulator and the deepest soil-moisture charge of the year. St. George stays active — citrus and cool greens keep producing.

DO THIS MONTH

CEDAR

Plan the year; read seed catalogs; sharpen tools; prune dormant trees above 28°F.

ST. GEORGE

Harvest greens, citrus, herbs; plant garlic and shallots; bare-root window opens.

BOTH

Order seed early — top varieties sell out by January.

BOTH

Build or refresh worm bins for spring inoculation.

QUICK WIN

Reflect and annotate this almanac — every December decision shapes May.

AVOID

Impulse-buying from glossy catalogs — stay regional.

WATCH FOR

Keep off frozen, snow-covered beds.

VERMIGOLD THIS MONTH — MAINTENANCE PRIMER



Both regions: a light dormant-season feed sustains the biology you built all year. 1:128 on a mild day, then plan next year’s full schedule.

PART FOUR

Annual Cycle at a Glance

MONTH	CEDAR FOCUS	ST. GEORGE FOCUS	VERMIGOLD
Jan	Plan / prune	Plant cool crops	Primer feed
Feb	Start indoors	First plantings	SG: First drench
Mar	Wake up	Heavy planting	Foundation drench
Apr	Spring sowing	Mulch heavy	Transplant soak
May	Warm-season in	Last warm window	Foliar spray
Jun	Peak growth	Pre-dormancy prep	Mid-month drench
Jul	Defend & harvest	Survival mode	Light foliar only
Aug	Begin fall sowing	Plan fall season	Recovery drench
Sep	Fall planting	Heavy fall planting	Major fall drench
Oct	First frost / garlic	Continue production	Cedar: Final drench
Nov	Mulch / shut down	First frost in SG	Both: Maintenance
Dec	Plan / dormant	Citrus / greens	Primer feed

Two big drenches

March and September do most of the system's work.

Offset calendars

One region is busiest when the other rests.

Year-round biology

Even the quiet months get a light primer feed — the workforce is never left untended.



V

PART FIVE

Soil Problem Diagnosis

*Before you treat anything, learn to read the ground. A
symptom is rarely the problem.*

PART FIVE

How to Read Your Soil

THE FIVE-MINUTE READ

Color. Rich brown is alive; pale gray-tan or red-orange means low organic matter.

Smell. Earthy and sweet means biology is alive; sour or odorless means something's off.

Structure. A moist handful should crumble into pea-sized aggregates — not a clay ball, not dust.

Life. Dig a 6" hole — healthy soil yields 2–5 earthworms and obvious life in 60 seconds.

Roots. White, branching, fine-haired roots are healthy; dark or J-shaped roots warn of compaction or salt.

THE SEVEN-QUESTION DIAGNOSIS

Water pools on the surface?	Crust / compaction
Water vanishes in seconds?	Sand / low OM
Yellow leaves, green veins?	Iron lockout
Wilts midday, recovers at night?	Shallow roots
Hard layer 6–18" down?	Caliche / hardpan
Same weeds in every bed?	Uniform soil
Different weeds across beds?	Microclimates

The right diagnosis saves you years. The next three pages are the most common problems — and their system fixes.

PART FIVE · PROBLEM ONE

Compaction, Crusting & Hardpan

SYMPTOMS

- ◆ Water pools and runs off instead of soaking in
- ◆ Hard to dig even when moist
- ◆ Stunted plants, shallow roots
- ◆ A crust forms within hours of watering
- ◆ A hard layer 6–18" below the surface

ROOT CAUSES

Compaction — traffic and tillage on wet soil destroy aggregate structure.

Crusting — bare soil + raindrop impact + high pH seals the surface.

Hardpan / caliche — natural calcium-carbonate layers at 12–24".

SYSTEM FIX — DO NOT TILL YOUR WAY OUT

- ◆ **Stop the cause.** Dedicated paths; no foot traffic; never work wet soil.
- ◆ **Cover the surface.** 3–4" mulch dissolves the crust within a season.
- ◆ **Build biology.** Vermicast, compost, and a VermiGold drench beat any tool.
- ◆ **Plant biological tillers.** Daikon, sunflower, alfalfa punch through compaction.

For hardpan: a one-time deep ripping (broadfork or single-tine plow) followed by aggressive biological rebuild can break a caliche layer permanently.

EXPECTED TIMELINE

Surface crusting fixes in one season. Compaction softens over two to three. Hardpan, once broken, stays broken if biology is maintained.

PART FIVE · PROBLEM TWO

Dry Soil & Drainage Failures

Both extremes — bone-dry and waterlogged — share one root cause: broken soil structure. Healthy soil holds water like a sponge, wet but never drowned.

SYMPTOMS — DRY

- ◆ Beds dry out within a day
- ◆ Midday wilt even with irrigation running
- ◆ Water beads up or runs off
- ◆ Dry an inch below the mulch

SYMPTOMS — WATERLOGGED

- ◆ Water pools after rain or irrigation
- ◆ Soil smells sour or sulfurous
- ◆ Roots dark, mushy, or shallow
- ◆ Unusually high disease pressure

SYSTEM FIX

- ◆ **Build organic matter.** Every 1% gain holds 16,000+ gallons per acre.
- ◆ **Mulch.** Cuts evaporation 50–70%.
- ◆ **Re-wet hydrophobic soil.** Non-chlorinated water + a wetting agent.
- ◆ **Shape the land.** Swales above dry zones, drainage away from soggy ones.

For salts: deep-flush two to three times a year with rainwater or filtered water — 4 to 6 inches in a single event.

EXPECTED TIMELINE

Mulching and biology fix evaporation in one season. Structural fixes — swales, organic-matter buildup — compound year over year.

PART FIVE · PROBLEM THREE

Poor Growth & Nutrient Lockout

SYMPTOMS

- ◆ Stunted despite adequate water
- ◆ Yellowing, especially new growth
- ◆ Bitter greens; flavorless tomatoes
- ◆ Fertilizer produces little response
- ◆ Tests read “adequate” but plants disagree

ROOT CAUSES

High pH locks micronutrients. At 8.0+, iron, zinc, manganese are present but unavailable.

Phosphorus tied by calcium. Only microbes free it economically.

Nitrogen starvation & salt buildup. Low OM and hard water mimic drought.

SYSTEM FIX

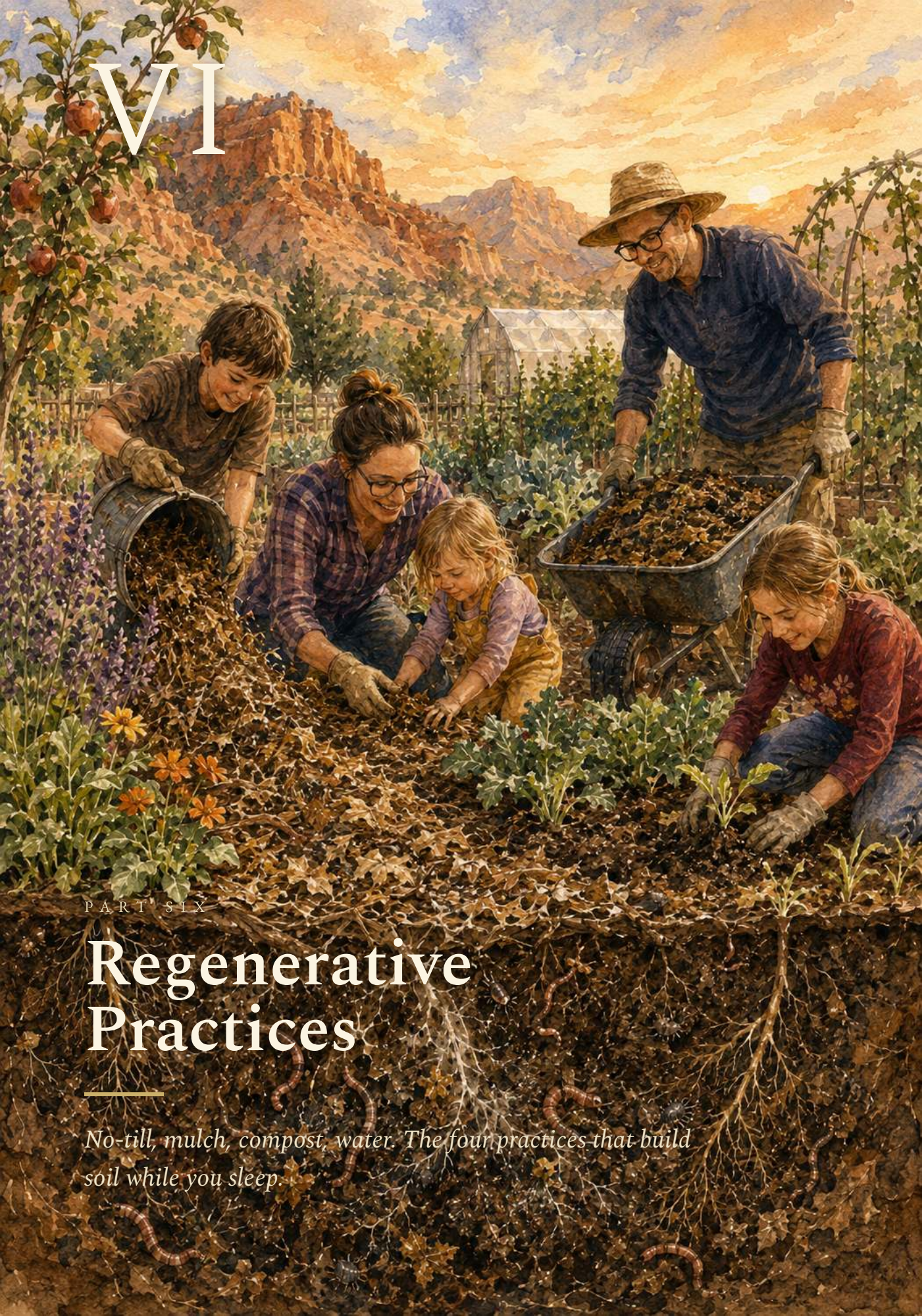
- ◆ **Drop pH biologically.** Microbial acids buffer the root zone — the highest-leverage move.
- ◆ **Inoculate with VermiGold.** Unlocks phosphorus and micronutrients within 30 days.
- ◆ **Build organic matter.** 4% OM is a 4× nitrogen reserve over bare desert soil.
- ◆ **Foliar-feed in critical windows.** Bypasses the root barrier when soil is locked out.

Use mineral acids — sulfur, gypsum, elemental sulfur — selectively, based on a soil test, never guesswork.

EXPECTED TIMELINE

Foliar feeding gives instant correction. Biological pH buffering shows in 30–90 days. Long-term lockout dissolves over two to three seasons of organic-matter buildup.

VI

A detailed illustration of a family of five working in a garden at sunset. A man in a straw hat and glasses stands with a wheelbarrow of mulch. A woman and two children are kneeling, spreading mulch over plants. A young girl is also kneeling nearby. The background features a greenhouse and a large, rocky mountain under a warm, orange sky. The foreground shows a cross-section of the soil with roots and worms.

PART SIX

Regenerative Practices

No-till, mulch, compost, water. The four practices that build soil while you sleep.

PART SIX

No-Till Foundations

Tillage is the single most destructive routine practice in conventional growing. It destroys aggregate structure, slices fungal networks, oxidizes carbon, and triggers a weed-seed explosion. Stopping it is the foundation of regenerative practice. The myth — “my soil is too hard to grow without tilling” — holds for one season; after two or three of no-till, biology, and cover, your soil is softer and deeper than any tilled bed.

HOW TO CONVERT

Sheet-mulch. Cardboard on the weeds, wet it, 4–6" compost + 3–4" straw. Plant into the compost.

Use a broadfork, not a tiller. It loosens without inverting layers.

Plant aggressively. Living roots break soil better than steel.

Top-dress, don't till in. Worms pull surface compost down.

WHAT IT DELIVERS

Water savings	30–60%
Weed pressure	–50% by yr 2
Earthworms	2–4×
Organic matter	+0.3–0.8%/yr

COMMON MISTAKES

Tilling “just one more time” to level it out · failing to mulch heavily (no-till requires cover) · expecting year-one yields to match a tilled bed — they won't, but year three exceeds it permanently.

No-till is the compounding interest of growing. Start now and the curve bends in your favor.

PART SIX

Mulching & Cover Crops

Mulch is not decoration — it is climate control.

MULCH TYPES FOR THE HIGH DESERT

Straw / oat	Cheap, fast — best for annual veg beds. Replace yearly.
Wood chips	Slow — best for trees, perennials, paths. 4–6".
Shredded leaves	Free in October. Mow once — excellent universal mulch.
Alfalfa hay	Adds nitrogen — best for heavy feeders.
Pine / juniper	Slightly acidifying — a benefit in our high-pH soils.

RECOMMENDED COVER-CROP BLENDS

WINTER · PLANT SEP–OCT

Cereal rye + winter peas + hairy vetch. Survives
Cedar winter; terminate in April.

SUMMER CEDAR · PLANT MAY

Buckwheat + crimson clover. Fast biomass, attracts
pollinators.

SUMMER ST. GEORGE · PLANT JUN

Cowpea + sorghum-sudangrass. Heat-loving,
deep-rooted, drought-tolerant.

QUICK ROTATION · ANY TIME

Daikon + clover + oats. Punches compaction, fixes
nitrogen in 60 days.

Terminate by mowing or rolling — lay residue on top, never till it in. Plant directly into the residue, allowing 2–3 weeks before warm-season planting.

PART SIX

Composting & Vermiculture

The desert breaks piles fast through evaporation and slow through cold. The right design accounts for both — and worm castings are the highest-value amendment per pound that exists.

COMPOSTING

Three-bin system, each 3'×3'×3' — one filling, one cooking, one finished.

Browns to greens, ~25:1 — about 2 parts dry browns to 1 part fresh greens.

Moisture: a wrung-out sponge; cover with a tarp to slow evaporation.

Turn once at week 2, once at week 4. 3–6 months in summer, 6–9 in winter.

VERMICULTURE

Two stacked totes, drilled for drainage and ventilation.

1 lb red wigglers (*Eisenia fetida*) — not standard earthworms.

Bedding: shredded cardboard, coir, or aged manure. Feed 1–2 lb scraps per lb of worms weekly.

Harvest every 3–4 months. Keep at 50–80°F; shade in summer, insulate in Cedar.

APPLICATION OF COMPOST & CASTINGS

Top-dress ¼–½" on every bed in March and September · worm castings 1–2 cups per square foot, twice a year
· brewed casting tea at 1:10 as a foliar or soil drench.

VermiGold liquid is essentially industrial-grade worm-casting extract — the same biology, applied at field scale.

PART SIX

Water Strategy for the High Desert

The water hierarchy — apply in order; each is more leveraged than the next.

- 1 Build soil that holds water.** 1% organic matter = 16,000+ extra gallons per acre. Compost, vermicast, mulch, cover.
- 2 Cover the soil.** A 2" mulch layer cuts evaporation 50–70%. This is free.
- 3 Capture every drop.** A 1,000 sq ft roof + 1" rain = 600 gallons. Swales on contour; rain gardens at downspouts.
- 4 Irrigate efficiently.** Subsurface drip beats surface beats overhead. Run pre-dawn; deep and infrequent trains roots down.
- 5 Match plant to water.** Natives, Mediterranean herbs, and adapted cultivars beat thirsty crops every time.

PRACTICAL ANNUAL TARGETS

CEDAR 12–18" irrigation / yr	ST. GEORGE 24–36" irrigation / yr	CONVENTIONAL SG 60–90" the gap is your savings
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Water is the most expensive input in our region. Every inch you save by building soil is money in the bank.

VII

PART SEVEN

Visual Charts & Tables

*The reference pages — planting calendars, soil temperatures,
and the numbers that decide everything.*

PART SEVEN

Regional Planting Calendar

● Direct sow ○ Transplant
▲ Indoor start ■ Harvest

CEDAR CITY / ENOCH · ZONE 6A–6B

	J	F	M	A	M	J	J	A	S	O	N	D
Peas			●	●		■	■		●			
Spinach			●	●		■		●	■	■		
Lettuce			▲	●	○	■		●	■	■		
Brassicas		▲	▲	○		■	■	●	○	■		
Carrots				●	●		■	■	●	■		
Beans					●		■	■	■			
Tomato			▲	▲	○		■	■	■	■		
Squash					●		■	■	■	■		
Garlic							■			●		

ST. GEORGE · ZONE 8A–8B

	J	F	M	A	M	J	J	A	S	O	N	D
Peas	●	●		■	■				●	■		
Lettuce	■	●	■						●	■	■	■
Brassicas	■	●	○					●	▲	○	■	■
Carrots	●	●		■	■				●	●	■	■
Beans		●			■	■			●	■		
Tomato	▲	○		■	■	■		▲	○	■		
Squash		●		■	■				●	■	■	
Garlic	■									●	●	

These are starting points. Annotate and adjust to your own microclimate.

PART SEVEN

Soil Temperature vs Crop Growth

A \$15 soil thermometer is the most useful — and most ignored — tool in the high desert. Plant by soil temperature, not by date.

MINIMUM GERMINATION TEMPERATURE

CROP	MIN	OPTIMAL
Spinach	35°	50–65°
Peas	40°	50–70°
Lettuce	40°	55–70°
Onion / Leek	35°	60–80°
Brassicas	40°	55–75°
Carrot	45°	60–75°
Beet	50°	60–80°
Bean	60°	70–85°
Corn	60°	75–85°
Tomato	60°	70–85°
Pepper	65°	75–85°
Squash	65°	75–90°
Melon	70°	80–95°

CRITICAL THRESHOLDS

32°F	Frost damage on tender crops
40°F	Cool-season root activity begins
50°F	Microbial activity rebounds
60°F	Warm-season window opens
70°F	Peak microbial efficiency
90°F	Plant heat stress begins
110°F	Photosynthesis shuts down
130°F	Soil microbes begin dying
150°F	Bare desert surface in July

READING TIP

Insert to 2" deep, same time daily. Mulched soil reads 5–15°F different — always check both.

PART SEVEN

Water Retention Improvement

How much water your soil holds at different organic-matter levels — and how fast you can climb.

ORGANIC MATTER	WATER HELD (GAL/ACRE/IN)	DAYS BETWEEN
0.5% — bare desert	~10,000	2
1.0% — untreated garden	~16,000	3
2.0%	~26,000	4–5
3.0%	~37,000	5–7
4.0% — regenerative target	~48,000	7–10
5.0%	~58,000	10–14

HOW TO RAISE ORGANIC MATTER

Bare cropping	–0.1 to –0.3%
Cover crops alone	+0.1–0.2%
Compost + mulch	+0.2–0.4%
+ cover crops	+0.3–0.6%
Full G.R.O.W.T.H.	+0.4–0.8%

YEARS TO REACH 4% (FROM 0.5%)

Conventional	never
Compost + mulch	8–12 yr
+ cover + biology	5–7 yr
Full + VermiGold	3–5 yr

Each 1% gain also buys ~50–70 lb of nitrogen reserve and 2–4× microbial biomass per acre.

Build organic matter, and your water bill, fertilizer bill, and yield curves all bend in your favor at once.

PART SEVEN

The Annual Application Wheel

The two big drenches

March (foundation) and September (fall recovery) do 60% of the system's work.

Mid-season touchpoints

April (transplant), May (foliar), June (mid-season) reinforce momentum.

The recovery drench

August, after the heat — restarts biology after high-temperature stress.

The quiet months

December–January in Cedar; July in St. George. Rest is part of the cycle.

The calendar anchors

Mulch, plant, harvest, cover crop — anchored to a season, not a date.

THREE CONSECUTIVE YEARS OF THE WHEEL

2–3×

organic matter

30–50%

lower water bill

30–80%

yield increase

near 0

synthetic dependence

A piece of land that compounds in value, not depreciates. The wheel turns the same direction every year.

VIII

PART EIGHT

The 12-Month Soil Transformation

Rebuild, Activate, Thrive. One year that changes the ground beneath your feet.

PART EIGHT

The Promise — Three Stages

Twelve months from today, you will look at the same piece of ground and not recognize it. A weak plant is a symptom. A pale leaf is a symptom. The cause is six inches below all of them.

Rebuild

MONTHS 1–3

Wake the dead. Bring the workforce back.

Activate

MONTHS 4–6

Lock in the engine. Plants stop needing rescue.

Thrive

MONTHS 7–12

Hand the work to the soil. The land starts paying you.

IF YOU FOLLOW IT

By month three the soil looks different; by six the plants do; by twelve you stop fertilizing and yields keep climbing. The water bill halves.

IF YOU DON'T

You spend another year buying nutrients you could be growing for free — watering more, harvesting less, losing ground.

Most growers try to fix their plants. The ones who succeed fix their soil.

STAGE ONE · MONTHS 1–3

Rebuild

*Wake the dead. Bring the
workforce back.*

WHERE YOU START

- ◆ Soil that crusts when wet, powders when dry
- ◆ A spade that hits hardpan at six inches
- ◆ Beds that need water every other day in May
- ◆ Worms you have to dig hard to find
- ◆ Plants that look tired for no clear reason

UNDERGROUND

The biology has been starving for years. Bacteria are scarce, fungi nearly gone — the food web is missing 80% of its workforce, and roots are doing the work of a whole ecosystem alone, and losing. Rebuild does one thing: it brings the workforce back.

WHAT CHANGES IN THE FIRST 30–60 DAYS

- ◆ The soil darkens half a shade as carbon accumulates.
- ◆ The smell turns from sour or odorless to earthy and sweet.
- ◆ Surface crusts soften under the first mulch layer.
- ◆ Water absorbs in seconds where it used to bead and run off.
- ◆ Worms appear in beds where there were none.
- ◆ New plantings respond visibly faster than ever before.

These are not yield gains yet — they are system signals. The soil is telling you it is waking up. Stop here and the biology starves again by August. Continue, and every stage compounds.

STAGE TWO · MONTHS 4–6

Activate

The soil is alive. Now put it to work.

WHAT YOU START TO SEE

Roots expand	Pull a finished crop — roots run two to three times deeper than in conventional soil.
Nutrient cycling	Yellowing disappears without fertilizer; locked phosphorus and minerals are unlocked.
Plants need less	Less water, less feeding, less intervention — the system does the work you used to.
Heat tolerance climbs	A bed that wilted at 95°F now holds steady at 100°F.
Pest pressure flattens	Aphids, leafminers, and the usual problems drop without sprays.

UNDERGROUND

Mycorrhizal fungi have networked with most of your roots, extending each plant's effective root system by an order of magnitude. Bacteria mineralize nutrients on demand; protozoa cycle nitrogen at the root surface. The food web is no longer scarce — it is functional.

This is where most growers realize it's working — and stop wondering whether the system works. The grower who reaches Activate does not go back.

STAGE THREE · MONTHS 7–12

Thrive

The climb is over. Now it compounds.

WHAT THE SOIL DOES FOR YOU

It feeds the plants

Growers running the full system typically step off conventional NPK by month nine.

It holds water

Beds stay moist 3–7 days longer between irrigations; water bills typically drop 30–50%.

It buffers heat

Surfaces stay 30–50°F cooler under cover; plants ride out heat domes that killed them in year one.

It builds itself

Aggregates form, hardpan softens, organic matter climbs 0.5–1% in twelve months.

It defends

Disease pressure drops; pest outbreaks become rare events, not seasonal expectations.

WHAT THE GROWER DOES

Less — significantly less. Applications get smaller and more strategic. Mulch breaks down on schedule, cover crops germinate where you laid them, and the weeds give way because the soil is doing what soil is supposed to do.

Most growers never reach this stage — because they stop too early. The one who runs the full twelve months owns a fundamentally different piece of land.

PART EIGHT

How to Execute the System

The three stages are the destination. This is the route — the execution rhythm, scaled to a half-acre.

MONTH	STAGE	APPLICATION	PURPOSE
Mar	Rebuild	Foundation drench	Wake the biology
Apr	Rebuild	Transplant soak	Establish young roots
May	Activate	Foliar spray	Build leaf defense
Jun	Activate	Mid-season drench	Sustain peak growth
Jul	Activate	Light foliar mist	Heat-stress relief
Aug	Thrive	Recovery drench	Restart after heat
Sep	Thrive	Major fall drench	The year’s key feed
Oct	Thrive	Transition drench	Lock in pre-winter
Nov	Thrive	Maintenance feed	Cool-season support
Dec–Feb	Thrive	Maintenance primer	Keep the workforce alive

Annual concentrate volume: **18–22 gallons per half-acre**; backyard plots (1,000–4,000 sq ft) run 4–8 gallons a year. Two heavy applications — March and September — do most of the work; the rest sustain momentum and keep biology alive year-round.

FOUR EXECUTION RULES

- ◆ Less, more often beats heavy and infrequent.
- ◆ Use non-chlorinated water — chlorine kills the biology.
- ◆ Apply on cool, calm days — sunrise or sunset.
- ◆ Combine with mulch and cover — biology needs food and shelter.

PART EIGHT

Application Methods

1 · SOIL DRENCH

1:32 · 4 OZ / GAL

The foundational application — biology to the root zone. 1 quart per 25 sq ft, on moist soil, every major application month. Watering can, drip injector, or coarse backpack sprayer.

2 · FOLIAR SPRAY

1:64 · 2 OZ / GAL

Biology onto the leaf surface — builds external defense and bypasses high-pH root lockout. Fine mist to slight runoff, sunrise or sunset only. Never in direct sun.

3 · TRANSPLANT SOAK

1:32 · 4 OZ / GAL

Establishes young plants with their full microbial partnership from day one. Submerge root ball or seed 5–30 minutes before planting, then a soil drench around the new plant.

A fourth option — dry vermicast top-dress. Apply ¼" of dry vermicast around active plants in March, June, and September; water in lightly. Pair with liquid VermiGold for full effect.

TOOL CARE

Rinse sprayers and tanks with non-chlorinated water immediately after use. Never use a sprayer that has held synthetic herbicide — residue kills biology. Store concentrate at 40–75°F, out of direct sun.

PART EIGHT

Recipes, Ratios & Storage

STANDARD DILUTION RATIOS

USE	RATIO	PER GAL
Heavy drench	1:32	4 oz
Standard drench	1:64	2 oz
Foliar spray	1:64–128	1–2 oz
Heat-stress mist	1:128	1 oz
Transplant soak	1:32	4 oz
Maintenance primer	1:128	1 oz

STORAGE

Unopened: 12 months at 40–75°F

Opened: 6 months refrigerated

Diluted: apply within 4 hours

Never freeze; never expose to chlorine

CUSTOM BLENDS

Spring Wake-Up · March

4 oz VermiGold + 1 cup molasses + 1 oz kelp, per gallon. Apply as a drench.

Summer Stress Foliar · Jul–Aug

2 oz VermiGold + 1 oz fish hydrolysate + 1 oz kelp, per gallon. Apply at sunset.

Fall Recharge Drench · September

6 oz VermiGold + 1 cup compost tea, per gallon.
Every active bed.

Water source: harvested rain, well, or carbon-filtered tap. Tap left open 24 hrs off-gasses chlorine — but chloramine must be neutralized.

The biology in the bottle is alive. Treat it like livestock, not chemistry.

THE FOUNDER'S STORY

The Hole That Changed Everything

JEFFREY JENSON · FOUNDER, VERMIGOLD SOIL SYSTEMS

The sun wasn't anything unusual that day. If you've ever worked outside in Southern Utah, you know the drill. Drink water. Keep moving. Finish the job.

At the time, I owned a sprinkler company. Long days in the heat were just part of life. But something happened that afternoon. **I got heat exhaustion.**

Most people recover, take a few days off, and move on. I never really did. At first it was hard to explain — I just wasn't myself anymore.

I was tired. Then exhausted. Then constantly exhausted. The weight started creeping on. The brain fog settled in. The aches became normal.

I kept telling myself I just needed to work harder. Sleep more. Push through. Drink another energy drink. Try another supplement. Eventually, I sold my company.

I spent a couple of years on the road figuring out what came next. When I came home, my wife and I were raising young children. I wanted to be present. I wanted to teach my kids useful things. So I started gardening — not because I thought it would change my life, but for some sunshine, some exercise, and maybe a few tomatoes.

Plants weren't new to me. My dad spent thirty-five years as a botanist with the Bureau of Land Management; some of my earliest memories involve plants, trees, and being outdoors.



THE JENSONS *Jeffrey and family, high desert of Southern Utah.*

But what I thought I knew about growing things was about to change.

When My World Got Small

In 2019, before COVID ever hit the headlines, my health fell off a cliff. I started experiencing severe digestive problems.

One episode landed me in a situation I'll never forget. For more than a week, I was losing blood. A lot of blood.

Doctors ran tests. Specialists ran tests. More appointments. More bills. More confusion. Thousands of dollars later, nobody could tell me exactly what was wrong.

That kind of uncertainty does something to a person.

For nearly a year, every trip away from home started with the same question: *"Where's the nearest bathroom?"*

A grocery run. A church meeting. A family outing. Everything revolved around that question.

My world got smaller and smaller. At one point, I was practically trapped inside my own house. I felt unsupported — bounced from doctor to doctor, specialist to specialist.

*Everyone had theories.
Nobody had answers.*

The Question That Wouldn't Leave

Eventually I stopped waiting for someone else to figure it out. I started researching. Books. Podcasts. Research papers. Nutrition. Gut health. Inflammation. Autoimmune disease. Anything I could find.

I discovered I had Hashimoto's. I discovered I had IBS. I discovered sugar and chocolate — two of my favorite things — were major triggers.

I experimented. Keto helped. Carnivore helped even more. Prolonged water fasting produced some of the biggest improvements I had seen in years. The more I simplified things, the better I seemed to feel.

Then I picked up a book called *The Dirt Cure*.

What if the health of our bodies begins with the health of the soil?

While I was trying to rebuild my health, I was also rebuilding my garden. I learned about compost. Regenerative agriculture. Johnson-Su bioreactors. Worms. Fungi. I learned that beneath every healthy plant exists an entire underground civilization most people never see.

Not Dirt. Soil.

One spring afternoon, I walked into the garden with a shovel. Three years earlier, the soil had been hard. Dry. Compacted. Lifeless — the kind of ground that fights back when you try to dig it.

This time was different. The shovel slid into the soil. When I lifted it out, the earth crumbled in my hand. Dark. Soft. Alive. Earthworms. Organic matter. Tiny roots everywhere. The smell reminded me of a forest after a rainstorm.

I stood there staring at a handful of soil. Not dirt. Soil. **Living soil.**

And suddenly something clicked. The same way our digestive system relies on bacteria and fungi to break down food, plants rely on them to break down organic matter. The same way diversity creates resilience in nature, diversity creates resilience in us. For years I had been focused on symptoms. The soil taught me to look at systems.

The vegetables tasted better. The garden became easier. The soil held water better. And I felt better eating food that came from living soil. I thought I was searching for a cure. What I was really searching for was a better way to live.

That's one of the reasons VermiGold exists — not because soil amendments are magic, but because healthy soil is where everything begins. Healthy plants. Healthy landscapes. Healthy communities. And maybe, just maybe, healthier people.

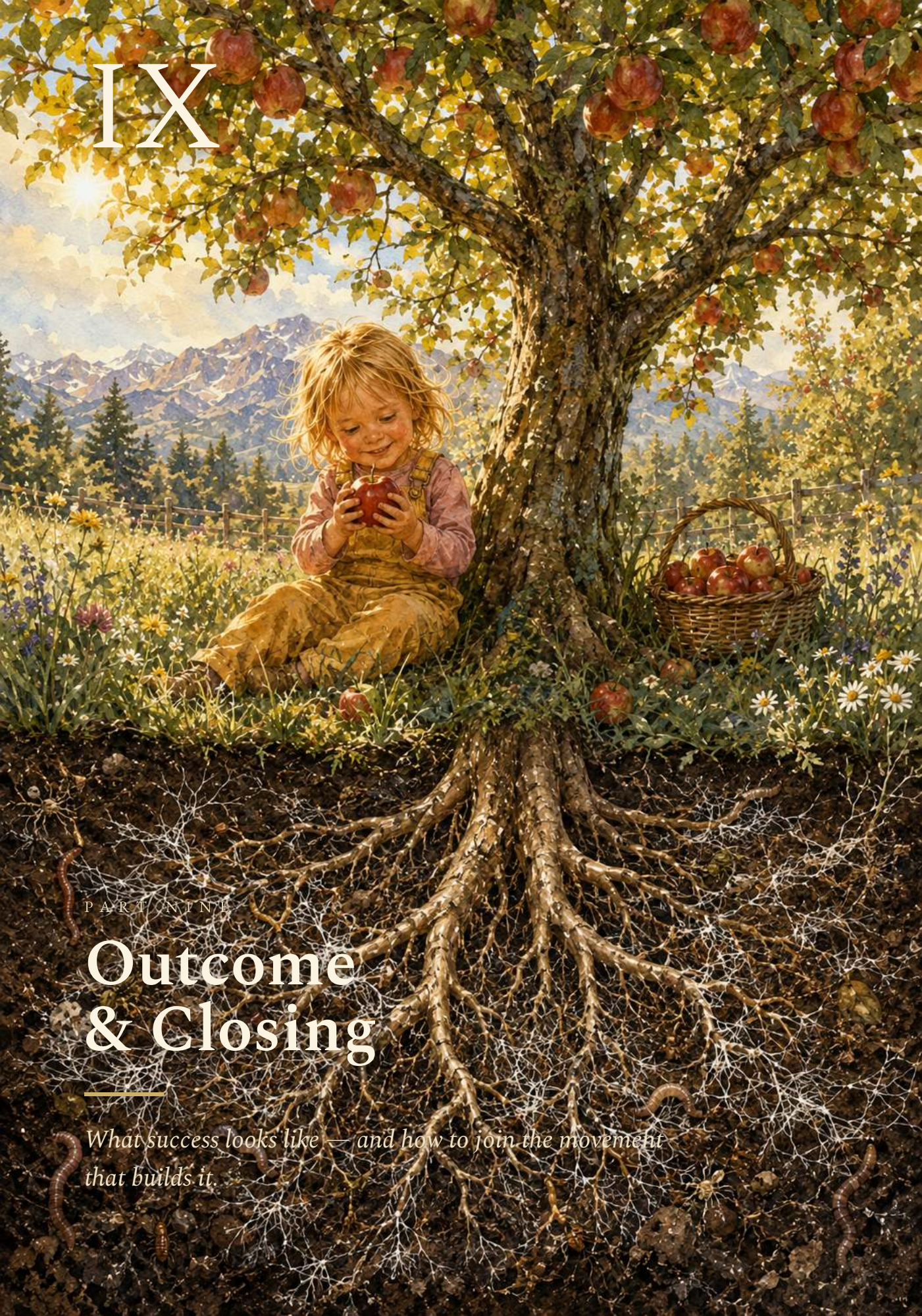


PLACEHOLDER a shovel of living soil —
art to come.

The hole I dug that day didn't solve all my problems. But it changed the questions I was asking — and sometimes the right question is worth more than the answer.

— Jeffrey Jenson

FOUNDER · CEDAR CITY, UTAH



IX

PART NINE

Outcome & Closing

*What success looks like — and how to join the movement
that builds it.*

PART NINE

What Success Looks Like

Typical of growers running the full G.R.O.W.T.H. System across our southern Utah field network. Results vary with starting soil, climate, and execution — the trajectory is what matters.

Year One

VISIBLE

Worms appear where there were none

Soil smells earthy and sweet

Mulched beds need water 30% less

Yields up 15–25%

Pest pressure lower by fall

Year Two

STRUCTURAL

Aggregate structure in every bed

Organic matter up 0.5–1.0%

Water bills drop 25–40%

Synthetic fertilizer cut or gone

Yields up 30–50%

Year Three

SYSTEMIC

Soil holds water for days

Handles 5–10°F more heat

Yields up 50–100%

Pest outbreaks become rare

Neighbors start asking

THE LONGER ARC

Five years in, the property is structurally different — hardpan softened, native pollinators returned, water needs cut 50–70%, yields 2× to 3× baseline. Ten years in, the land is a working ecosystem: self-fertilizing, mostly self-watering — a piece of high desert that has become high-functioning farmland.

Not a marketing promise — an observable, measurable, compounding outcome.

PART NINE

Joining the Soil Movement

The high desert is not a difficult place to grow food. It is a place that demands you grow soil first. You now have the framework, the regional intelligence, the playbook, and the tools. What's left is the doing.

THREE COMMITMENTS FOR THE YEAR AHEAD

- 1 Pick one principle.** Run one of the six G.R.O.W.T.H. letters relentlessly for a full year. Layer the next next year.
- 2 Keep a journal.** One observation a week. By 2030 it is the most accurate forecasting tool on your property.
- 3 Talk to your neighbors.** Every regenerative grower you connect with shortens your learning curve by years. The movement is local.

*Soil is the slowest-compounding asset on
Earth — and the most generous.*

The desert is patient. The grower who matches the desert's patience wins. The land is not against you — it has been waiting for someone to listen.

— The VermiGold Team

CEDAR CITY, UTAH · 2026



VermiGold

SOIL · LIFE · REGENERATION

*Healthy soil grows healthy plants.
Healthy plants heal the planet.*

The VermiGold G.R.O.W.T.H. System™ is a registered trademark of VermiGold Soil Systems. This almanac is regional in scope — the principles travel, the specifics are tuned for southern Utah.

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